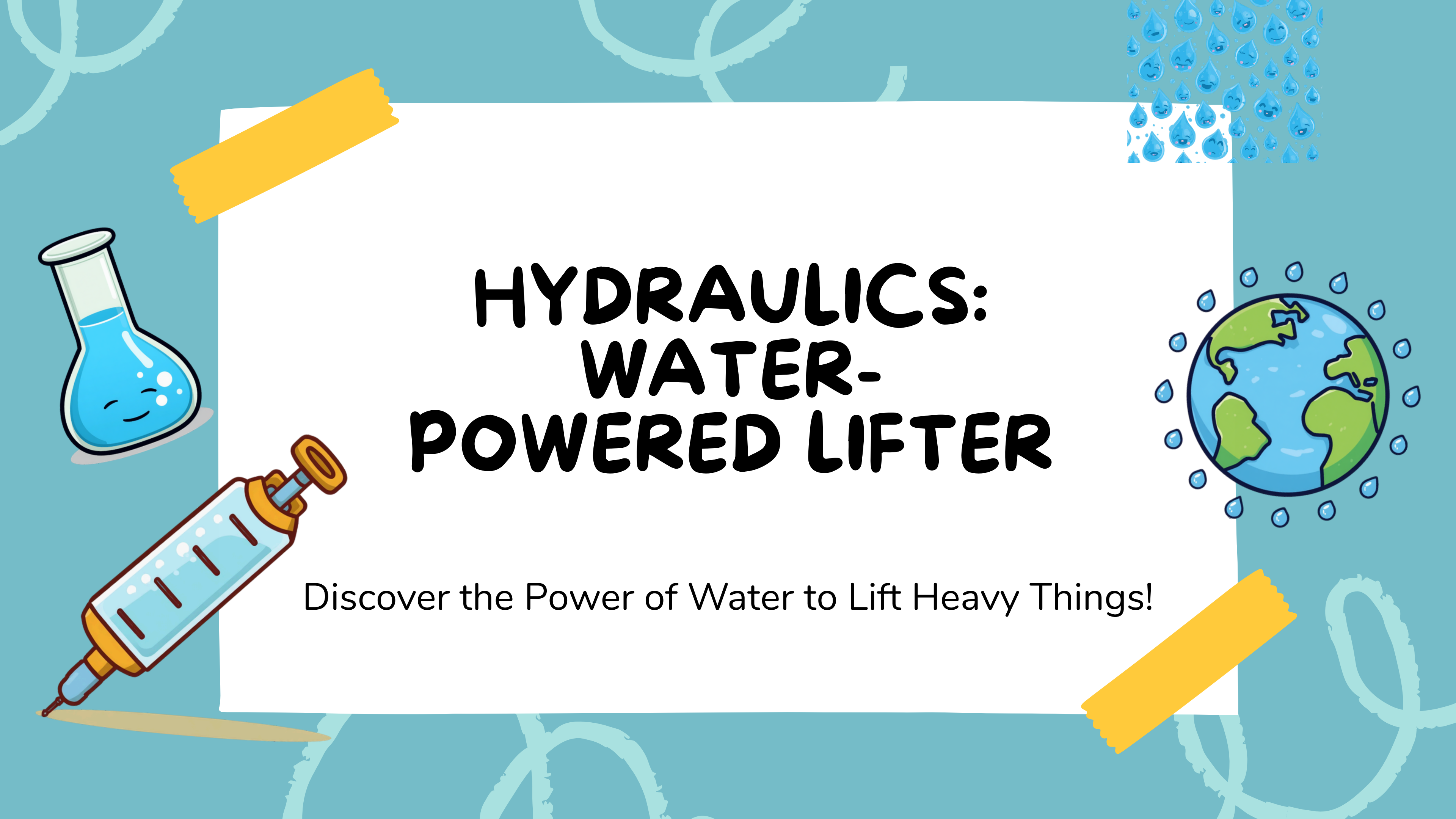




# HYDRAULICS: WATER- POWERED LIFTER

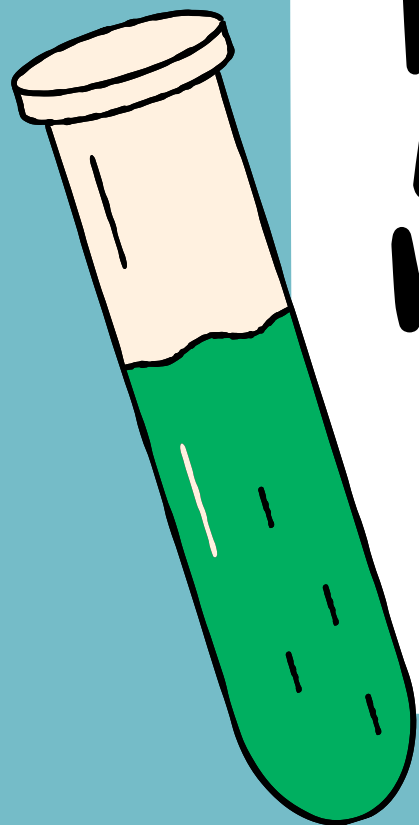
Discover the Power of Water to Lift Heavy Things!

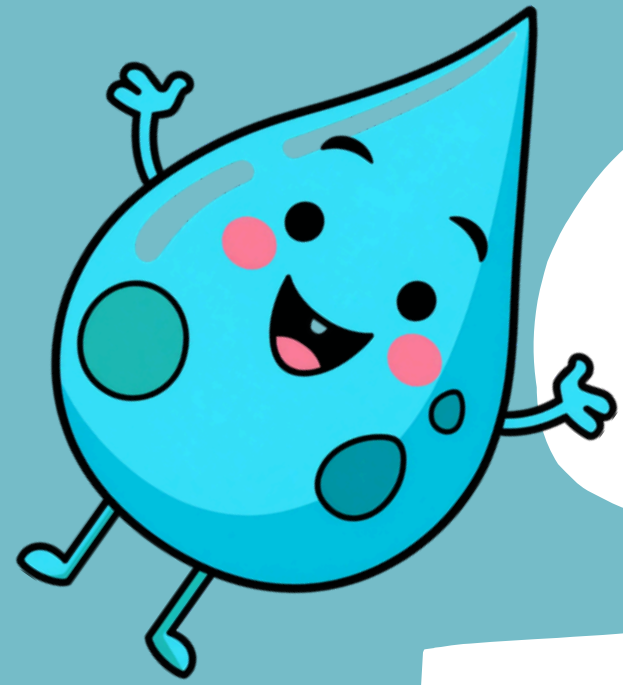






**How does this big  
machine lift such  
heavy things with  
water?**





# What is Hydraulics?

## Definition

Hydraulics means using water or other liquids to move or lift things.

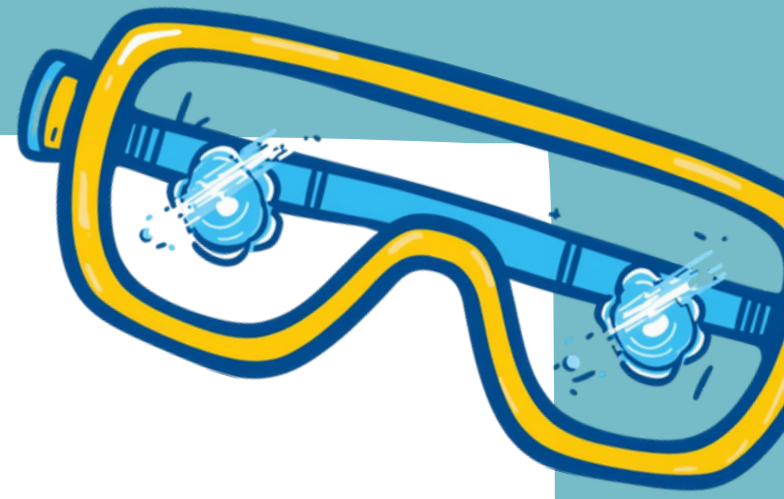
## Key Idea

Water helps us do big jobs with small pushes!





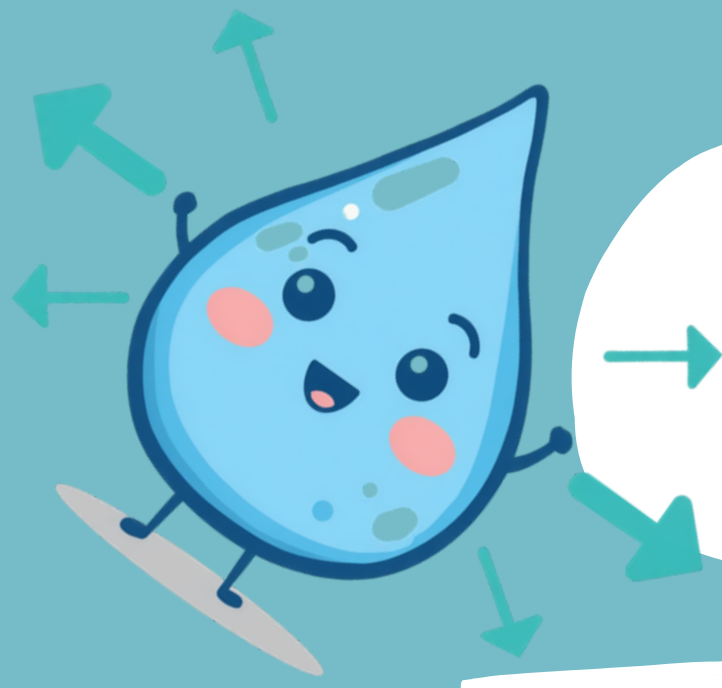
# Water vs. Air: The Science of Squish



You can squish a sponge but water does NOT get smaller. This is called Incompressibility!







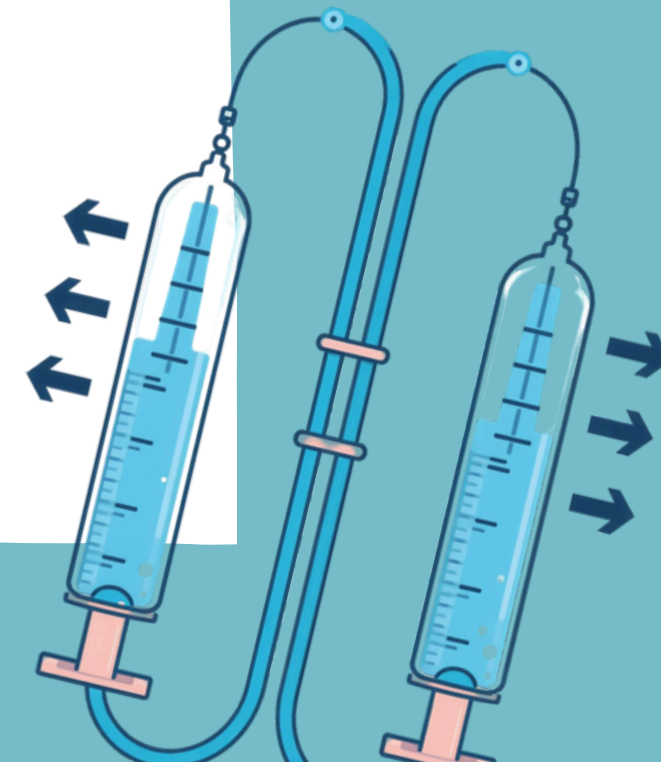
# Pascal's Law Made Easy

## The Rule

When you push water in a closed pipe, the push spreads everywhere equally!

## Example

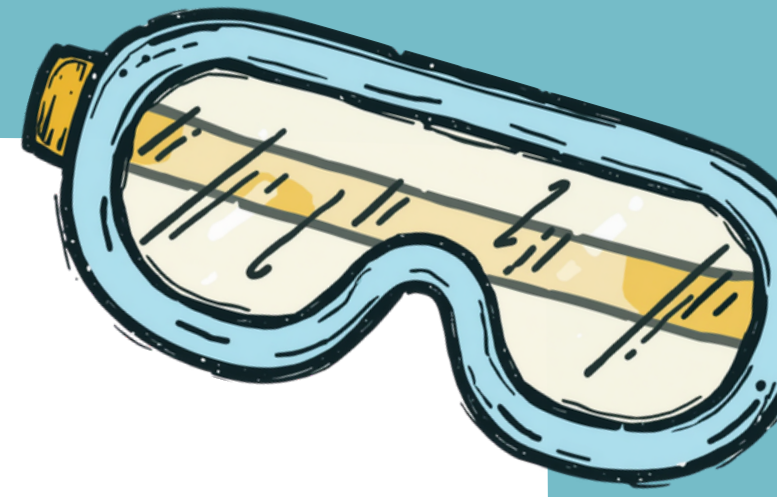
If you press one side of a water pipe, the other side feels the same push!



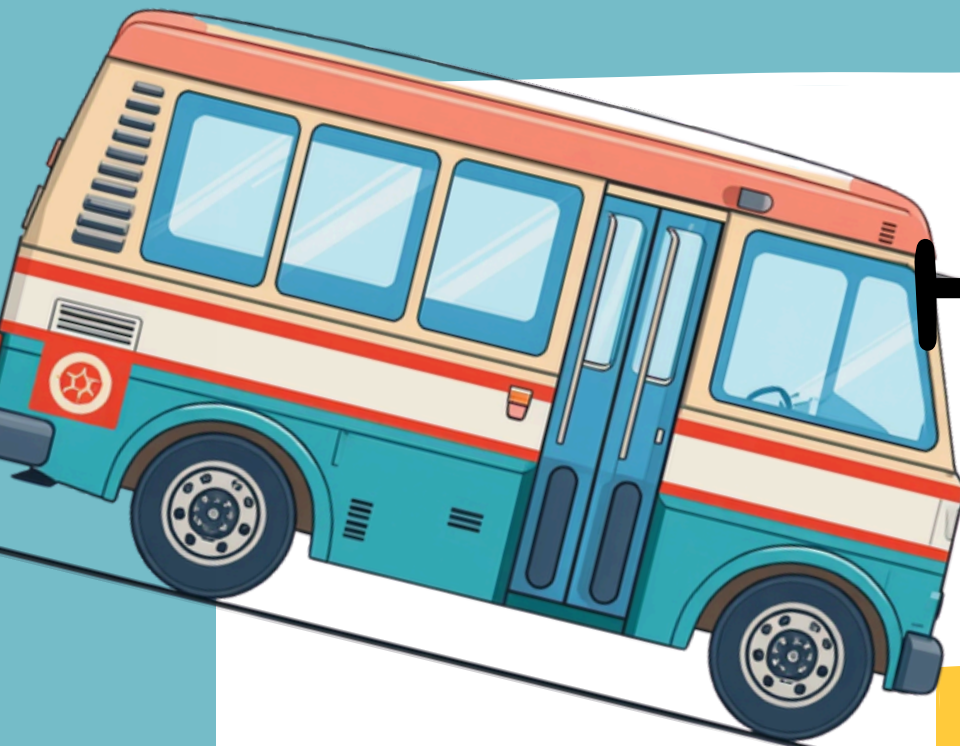
# Push and Pull with Syringes



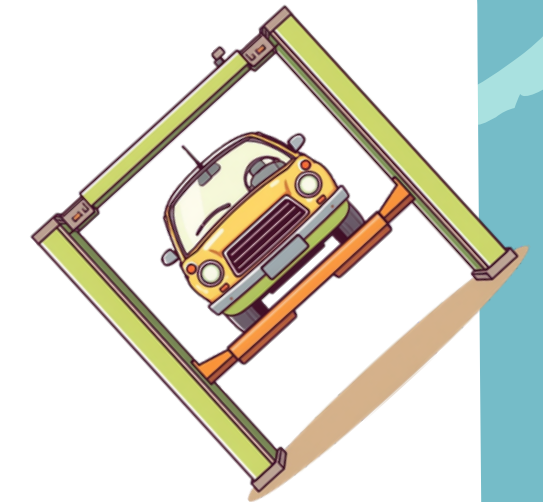
Push one syringe, and the other pushes out too. This is how we lift things!







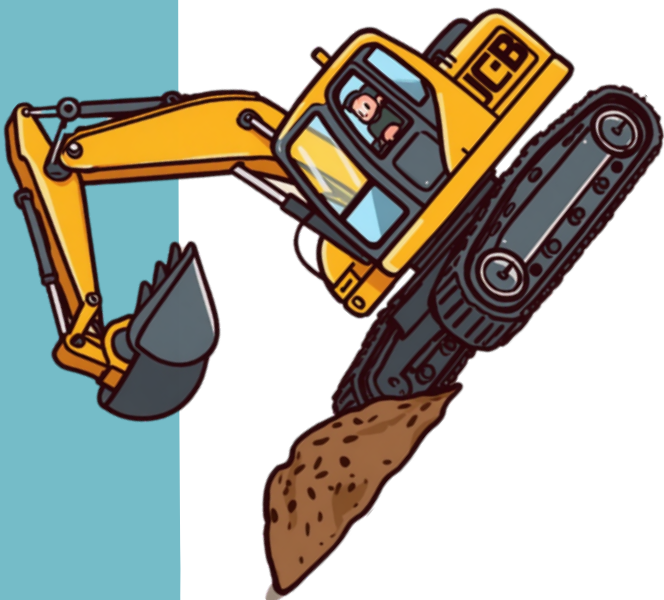
# Hydraulics in India Around Us



JCB at Construction Sites

Hydraulic Brakes on Buses

Car Lifting Platforms



JCB machines use water power to lift heavy rocks and dig soil at building sites!

Buses and trains use hydraulic brakes to stop safely. Water pressure helps slow down heavy vehicles!

# Let's Build Our Water-Powered Lifter!

1 Two syringes

2 Plastic tubing

3 Small platform

4 Water & cup

5 Connect syringes with  
tubing filled with water

6 Attach platform to one  
syringe's plunger

7 Push one syringe to  
lift the platform!

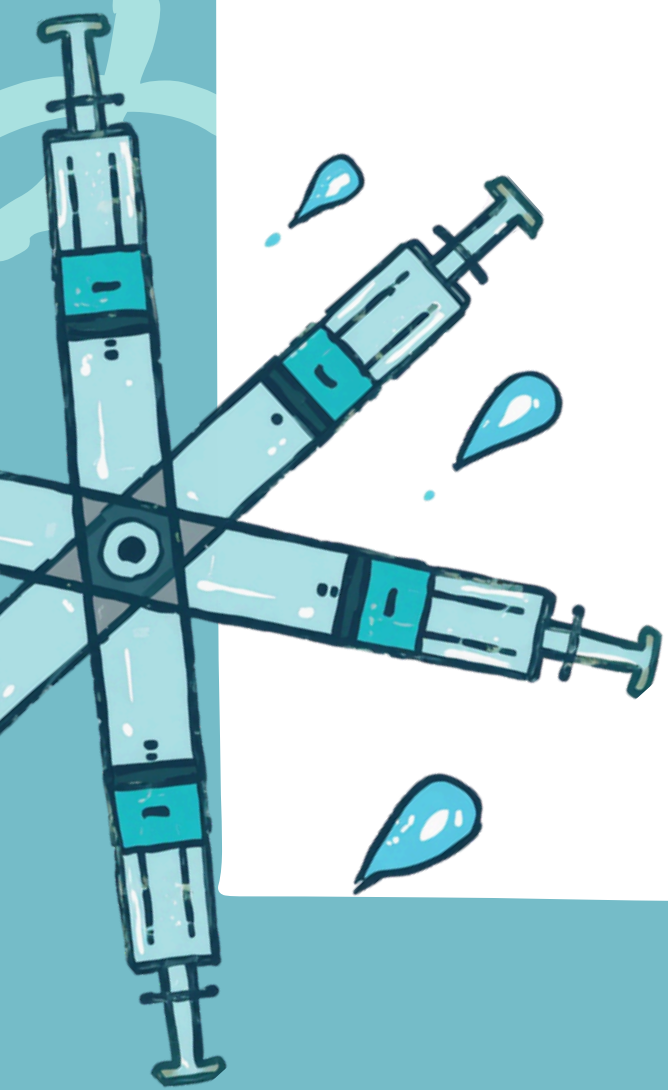
**You're the engineer now!**  
Start building and experimenting!

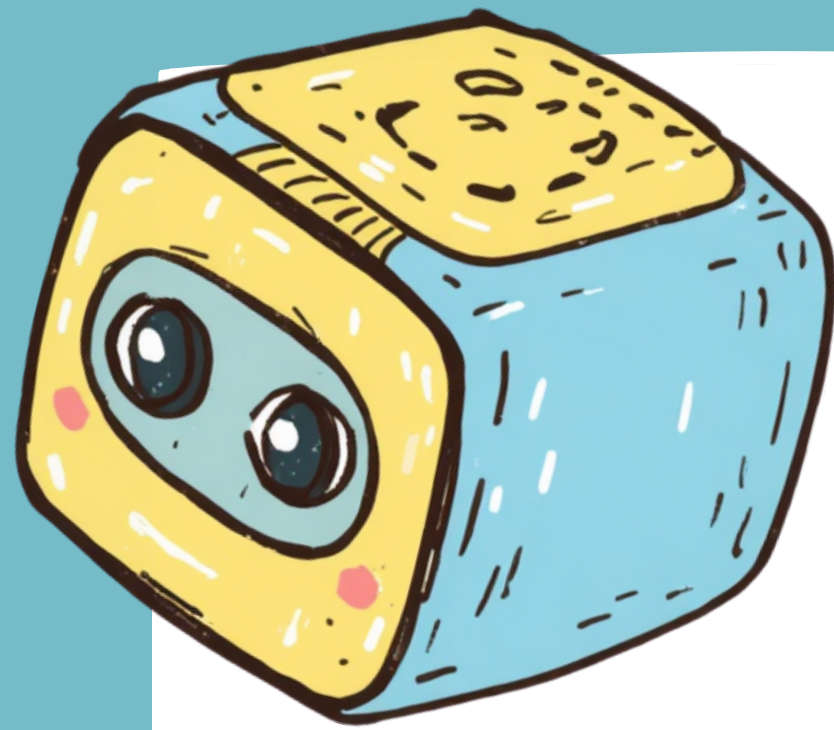


# Watch the Lifter in Action



Look how  
water lifts the  
platform when  
we push here!





# Challenge Time!



Eraser (Light)

Sharpener (Medium)

Heavy Stone (Hard)



Can your lifter pick up these objects? Try each one and see what happens!



Experiment and note which objects your hydraulic lifter can pick up. What works best?



# What Did We Learn?



Water cannot be  
squished - it's  
incompressible!



Push one syringe,  
water moves  
everywhere equally!

Small push can lift  
heavy objects with  
hydraulics!





## **Ready to be a Hydraulic Hero?**

Let's see how much you know! Answer these fun questions and show off your water power skills. You're becoming experts in the power of water!





# Questions & Fun Facts!

Did you know? The hydraulic lift can lift cars weighing thousands of kilos! Ask your questions now!







# Thank You & Keep Exploring!

THANK YOU FOR JOINING THE WATER-POWERED LIFTER ADVENTURE!  
KEEP EXPERIMENTING AND ASKING QUESTIONS. - PLANRS ACADEMY